

Conjecture 146: the stated normalization has no weak probability limit

Local development pilot; not an upstream submission

8 October 2026

The source uses primes $p \leq N$, denominator N , and scaling N . We preserve all three. Its supremum domain and law of α are unstated. We use the conventional Kolmogorov–Smirnov domain $[0, 1]$ and prove a stronger statement covering every probability law of α , even one changing with N . A supremum over all real thresholds would be unbounded; that interpretation is not used as a shortcut.

For $N \geq 1$, put $\mathcal{P}_N = \{p \in \mathbb{N} : p \text{ prime, } p \leq N\}$ and

$$F_N(\alpha, t) = \frac{\#\{p \in \mathcal{P}_N : \{\alpha p\} < t\}}{N}, \quad D_N(\alpha) = \sup_{0 \leq t \leq 1} |F_N(\alpha, t) - t|, \quad X_N(\alpha) = ND_N(\alpha).$$

The Lean definitions are total at $N = 0$ using real division by zero; every limiting argument only uses $N \geq 4$.

Theorem 1. *For any sequence of Borel probability measures μ_N on $\mathbb{R}$, the actual pushforward laws $\lambda_N = (X_N)_*\mu_N$ have no weak limit which is a probability measure on $\mathbb{R}$. In particular they cannot converge to the law of any real-valued Brownian-bridge functional.*

Proof. Every fractional part belongs to $[0, 1)$, so $F_N(\alpha, 1) = \#\mathcal{P}_N/N$ for every α . Hence

$$X_N(\alpha) \geq N - \#\mathcal{P}_N \geq N/2 - 2 \quad (N \geq 2). \tag{1}$$

For the last inequality, all primes other than 2 are odd. Counting those integers gives $\#\mathcal{P}_N \leq N/2 + 2$; no prime number theorem is needed. The Lean proof uses the corresponding arithmetic prime-counting bound.

These are genuine measurable statistics. For fixed t , each comparison $\{\alpha p\} < t$ is measurable: multiplication and fractional part are Borel measurable. A finite sum of indicators therefore makes $F_N(\cdot, t)$ measurable. Furthermore,

$$D_N(\alpha) = \sup_{q \in \mathbb{Q} \cap [0, 1]} |F_N(\alpha, q) - q|. \tag{2}$$

To check the nontrivial direction, take $t > 0$. The finite collection of samples strictly below t has a maximum below t , if nonempty. Choose rational $q < t$ arbitrarily close to t and above every such sample. Then the strict-threshold count at q equals the count at t , and the displayed errors approach one another. The case $t = 0$ is itself rational. This argument also handles $t = 1$, repeated samples and an empty collection. The ranges of errors are nonempty and bounded, as checked in Lean. Equation (2) is a countable supremum of measurable functions; thus D_N and X_N are measurable, and λ_N is indeed a probability measure.

Use the bounded continuous, everywhere strictly positive test

$$f(x) = \frac{1}{1 + |x|}, \quad 0 < f(x) \leq 1.$$

For $N \geq 4$, (1) implies

$$0 \leq \int f d\lambda_N = \int f(X_N(\alpha)) d\mu_N(\alpha) \leq \frac{1}{N/2 - 1} \rightarrow 0. \quad (3)$$

The equality is the proved pushforward integral identity. If λ_N converged weakly to a real probability measure ν , this bounded continuous test would also give $\int f d\lambda_N \rightarrow \int f d\nu$. Yet $\int f d\nu > 0$: a nonnegative measurable function has zero integral only if it vanishes almost everywhere, whereas f is positive everywhere and $\nu(\mathbb{R}) = 1$. This contradicts (3). $\square$

Formal correspondence. `Route2.primeKS` is the actual supremum over the subtype $[0, 1]$; `scaledKS` multiplies it by N . `measurable_primeKS` proves (2)’s measurability consequence, and `laws_probability` and `test_transport` certify the actual image laws. The zero-limit and strictly-positive-integral theorems supply the contradictory limits. The final theorem is `Route2.no_probability_weak_limit`. Its explicit weak-convergence predicate tests all continuous real functions with values in $[0, 1]$, using their nonnegative lower integrals. This is the usual normalized bounded-continuous-test formulation for probabilities; in particular standard weak convergence necessarily passes the test used above. No extra premise assumes measurability, the escape bound, or an object bridge.

Provenance and limits. The endpoint bound, rational-supremum equality and fixed-threshold CDF measurability reuse disclosed Phase 1c B partial work; an uncompiled Phase 1c A draft supplied the proposed integral route. This round completes the measurability, probability-law and weak-limit connections. Those prior ideas are not claimed as new discoveries. The exact original, pinned Lean sources, source correspondence and independent verification records accompany this local package. Local verification does not establish submission eligibility, a first solve, or upstream acceptance.